

Last time A seq. (a_n) is a Cauchy if $\forall \varepsilon > 0, \exists N$ s.t. whenever $m, n \geq N$ then $|a_m - a_n| < \varepsilon$.

Remark Sequential Criterion for uniform continuity:

A function f is unif. continuous if & only if for every Cauchy seq. (a_n) . We have that $(f(a_n))$ is Cauchy.

PF $(\Rightarrow)$ Assume f unif. cts. and let (a_n) be a Cauchy seq.

WTS. $\forall \varepsilon > 0$, want $N \geq 0$ s.t. whenever $m, n \geq N$

We have $|f(a_n) - f(a_m)| < \varepsilon$

Since f unif. cts. for a given $\varepsilon > 0$, $\exists \delta$ s.t. whenever $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

As long as $|a_n - a_m| < \delta$. then we will have $|f(a_n) - f(a_m)| < \varepsilon$.

Now let N be s.t. whenever $m, n \geq N$, $|a_n - a_m| < \delta$

And this N works to show that $(f(a_n))$ is Cauchy.

$(\Leftarrow)$. Assume (a_n) is Cauchy $\Rightarrow (f(a_n))$ Cauchy

WTS f unif. cts.

Given $\varepsilon > 0$, $\exists \delta > 0$, whenever $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$

We should prove this by contradiction. So assume f not unif. cts.

$\exists \varepsilon > 0$, s.t. $\forall \delta > 0$, can find x, y w/ $|x - y| < \delta$ but $|f(x) - f(y)| > \varepsilon$

Goal To find a Cauchy seq. (a_n) s.t. $(f(a_n))$ is not Cauchy.

$$\text{Let } \epsilon=1, \delta=1$$

$$\exists x, y \text{ s.t. } |x-y| < \delta \text{ but } |f(x)-f(y)| > 1$$

TBC on Thursday

LUB is an axiom on completeness.

(A system is complete if every Cauchy Seq. converges)

$\mathbb{R}$ = complete

$\mathbb{Q}$ = not complete

Instead of LUB property, we can also assume Nested Intervals Property
 Let $[a_0, b_0] \supseteq [a_1, b_1] \supseteq [a_2, b_2] \supseteq \dots$ be a seq. of nested closed
 point intervals. Assume $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$

Then $\bigcap_{n=0}^{\infty} [a_n, b_n]$ is a single element.

Thm. LUB $\Leftrightarrow$ NIP

Pf. ($\Leftarrow$) Assume NIP, prove LUB

Let $A \subseteq \mathbb{R}$ be a set, $A \neq \emptyset$, bounded above

A nonempty $\Rightarrow x_0 \in A$

A bounded $\Rightarrow M_0$, UB for A

Look at $\frac{x_0 + M_0}{2}$

If $\frac{x_0 + M_0}{2}$ is bigger than every elt of A

Let $x_1 = x_0$, $M_1 = \frac{x_0 + M_0}{2}$

$\uparrow$ still a UB for A

If $\frac{x_0 + M_0}{2}$ is not larger than every elt of A .

$$x_1 = \text{elt of } A \geq \frac{x_0 + M_0}{2}$$

$$M_1 = M_0$$

Now repeat on $[x_1, M_1]$ ← length $\leq \frac{M_0 - x_0}{2}$

$$\lim_{n \rightarrow \infty} (M_n - x_n) \leq \frac{M_0 - x_0}{2^n} \rightarrow 0$$

$$\text{NIP} \Rightarrow \bigcap [x_n, M_n] = \{\alpha\}.$$

Claim. α is the LUB of A

• α is an UB of A .

If not, $\exists x \in A, x > \alpha$

Let n be large enough s.t. $\frac{M_0 - x_0}{2^n} < \alpha - x$

• If any interval of length $< \frac{M_0 - x_0}{2^n}$ contains α , then by definition of n , it does not contain x .

$\Rightarrow x > M_n$, contradiction.

• α is the LUB of A If not, let β be a smaller UB of A .

Let n large enough s.t. $\frac{M_0 - x_0}{2^n} < \beta - \alpha$

Then if $[x_n, M_n] \ni \alpha$

then $\beta \notin [x_n, M_n]$

$\Rightarrow x_n > \beta$, so β is not an UB

$\Rightarrow \alpha$ is LUB of A .

$\text{LuB} \Rightarrow \text{NIP}$

$$[a_0, b_0] \supseteq [a_1, b_1] \supseteq \dots$$

$$\lim_{n \rightarrow \infty} (b_n - a_n) = 0$$

$$A = \{a_0, a_1, \dots\}$$

$A \neq \emptyset$, bounded above b_0

$$\Rightarrow \alpha = \text{LuB}(A)$$

Show that $\{\alpha\} = \bigcap [a_n, b_n]$

Need $\alpha \in [a_n, b_n] \forall n$

no other $\beta \in \bigcap [a_n, b_n]$.